

QUIZ APP SOFTWARE DEVELOPMENT

Week 4 Project Report

Question Management System

Project	Quiz App
Week	Week 4
Topic	Question Management System
Concepts	Arrays, ArrayList, Java Collections, Data Storage

1. Introduction

The Quiz App is a Java-based software project designed to provide a simple and interactive quiz experience. In Week 4, the project focuses on developing the Question Management System. The main objective is to create a question database, store questions and answers, display questions dynamically, and organize questions into quiz categories.

2. Week 4 Objectives

The main objectives of Week 4 are:

- Create question database.
- Store questions and answers.
- Display questions dynamically.
- Organize quiz categories.

3. Question Management System

The Question Management System is a core module of the Quiz App. It manages the questions and answers used during a quiz. The module provides a structured way to store questions, retrieve them when needed, display them to users, and organize them according to different categories.

4. Question Database

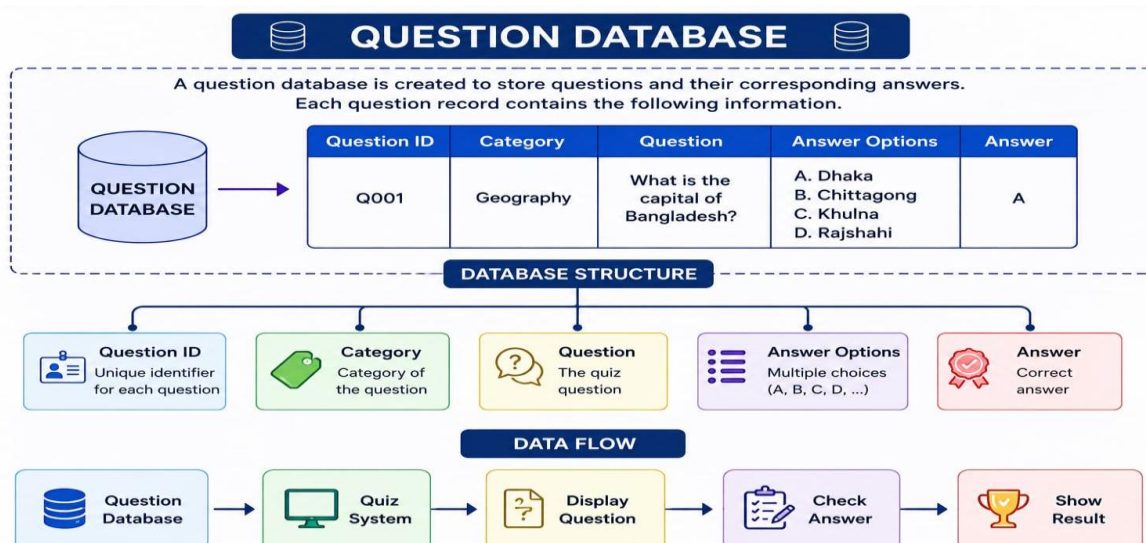
A question database is created to store questions and their corresponding answers. For the initial

implementation, Arrays or ArrayList can be used as an in-memory data store. Each question record contains

basic information required by the quiz system.

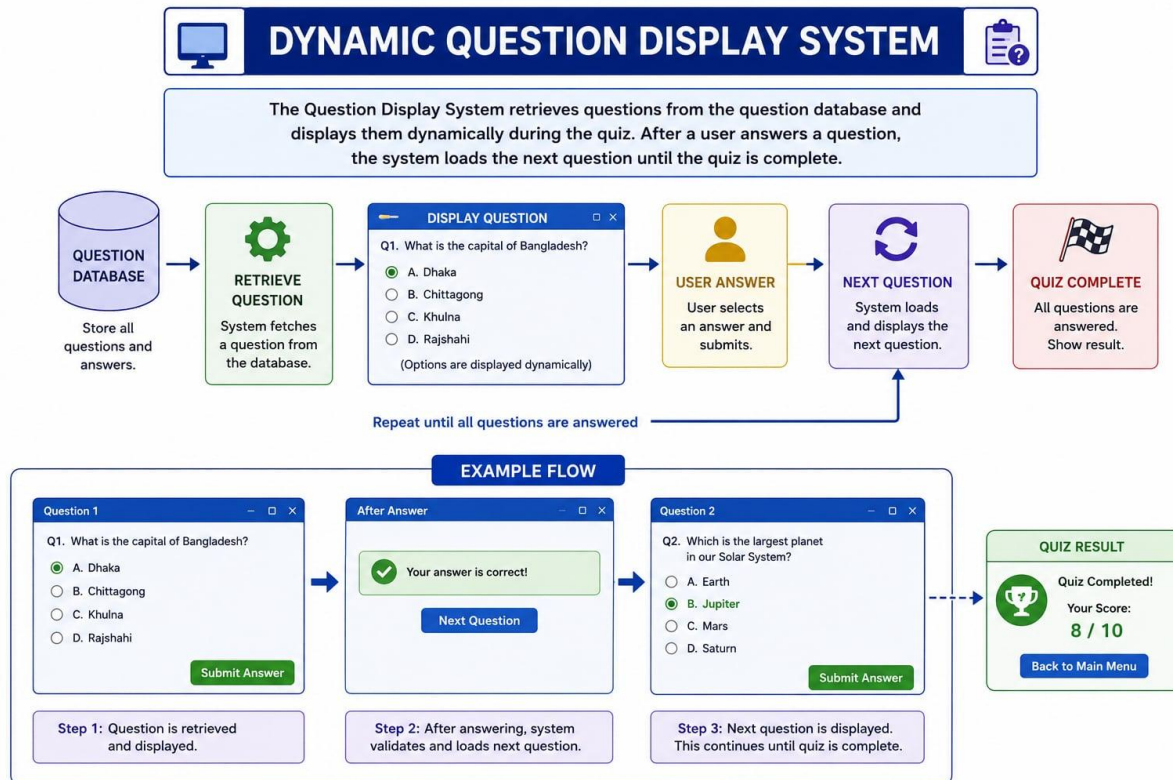
Example question information includes:

- Question ID
- Category
- Question
- Answer
- Answer Options



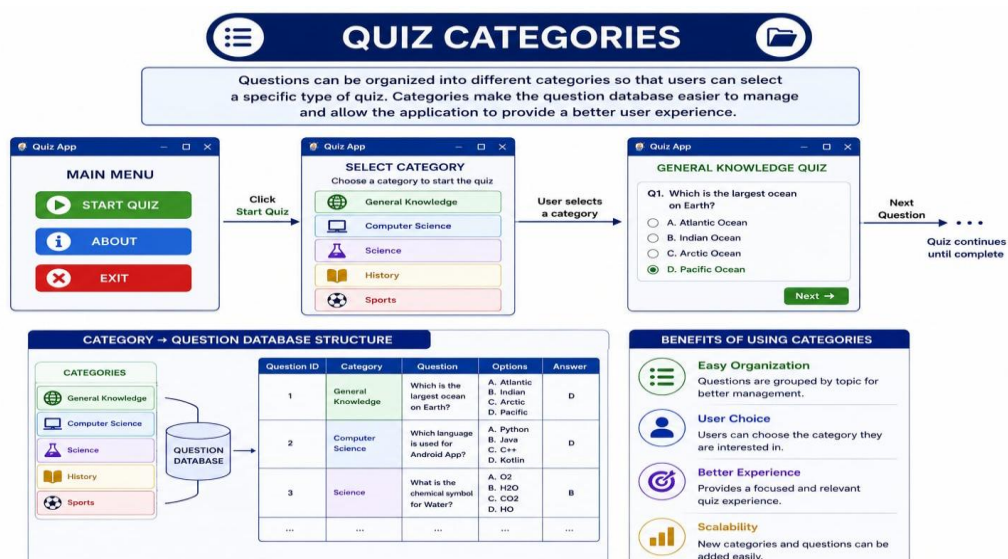
5. Dynamic Question Display

The Question Display System retrieves questions from the question database and displays them dynamically during the quiz. After a user answers a question, the system can load the next question until the quiz is completed.



6. Quiz Categories

Questions can be organized into different categories so that users can select a specific type of quiz. Categories make the question database easier to manage and allow the application to provide different quiz topics.



7. Technologies and Concepts to Learn

Technology / Concept	Purpose
Arrays and ArrayList	Store and manage multiple questions and answers.
Java Collections	Provide data structures and methods for managing question data.
Data Storage Concepts	Understand how question information is stored and retrieved.

8. GitHub Upload

The Week 4 implementation will be uploaded to the project GitHub repository. The required materials are:

- Question Module
- Question Database
- Related Java source files required to run the question module.

9. Demonstration

At the end of Week 4, the project should demonstrate a working Question Display System and basic Question Management. The application should retrieve questions from the database, display questions and answer options dynamically, and organize questions by category.

Demonstration Flow:

Select Category → Retrieve Question → Display Question → Select Answer → Load Next Question

10. Expected Outcome

After completing Week 4, the Quiz App will have a basic Question Management System. The application will be able to store questions and answers, display questions dynamically, and organize questions into different categories. This module will provide the foundation for implementing the complete quiz functionality in later weeks.

11. Conclusion

Week 4 focuses on developing the Question Management System of the Quiz App. By using Arrays, ArrayList, Java Collections, and data storage concepts, the project will be able to manage question data in an organized way. The completed module will support dynamic question display and category_based quiz organization.